Bayes' Theorem: an introduction

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Previously on ELE8088

Basic probability theory:

- Probability spaces
- Discrete and continuous random variables
- Expectation
- Variance
- Normal distribution (in one and several dimensions)
- Marginal distributions
- Conditioning (conditional probability, expectation and variance)
- Conditioning of multivariate normals

Bayes' Theorem

Conditional probability version

Bayes' Theorem

We have defined the conditional probability as

$$\mathsf{P}[A \mid B] = \frac{\mathsf{P}[A \cap B]}{\mathsf{P}[B]},$$

provided $\mathsf{P}[B] > 0$. Similarly,

$$\mathsf{P}[B \mid A] = \frac{\mathsf{P}[A \cap B]}{\mathsf{P}[A]},$$

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$$\boxed{\mathsf{P}[A \mid B] = \frac{\mathsf{P}[B \mid A]\mathsf{P}[A]}{\mathsf{P}[B]}} \tag{Bayes}$$

with $\mathsf{P}[B] > 0$ (it holds for $\mathsf{P}[A] = 0$).

Law of Total Probability

We have shown (Exercise in previous section) that for $A \in \mathcal{F}$ and a sequence of events $(B_n)_n$ that *partitions* Ω, it is

$$\mathsf{P}[A] = \sum_n \mathsf{P}[A \cap B_n].$$

Wlog we assume $\mathsf{P}[B_n] > 0$ for all n. Then $\mathsf{P}[A \cap B_n] = \mathsf{P}[A \mid B_n]\mathsf{P}[B_n]$, so

$$\boxed{\mathsf{P}[A] = \sum_n \mathsf{P}[A \mid B_n]\mathsf{P}[B_n].}$$

This is known as the <u>law of total probability</u> (LTP).

By LTP, for $A, B \in \mathcal{F}$,

$$\mathsf{P}[A] = \mathsf{P}[A \mid B]\mathsf{P}[B] + \mathsf{P}[A \mid B^c]\mathsf{P}[B^c].$$

Do I have COVID-19?

A certain test for COVID-19 has *sensitivity* 99%, that is

$$\mathsf{P}[\text{Positive} \mid \text{COVID}] = 0.99.$$

The test has *specificity* 98%, that is

$$\mathsf{P}[\text{Negative} \mid \text{Healthy}] = 0.98.$$

The *prevalence* of COVID is 10%, i.e.,

$$\mathsf{P}[\text{COVID}] = 0.1.$$

These probability values are arbitrary

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What is the probability that a random person tests positive? By the law of total probability:

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&= 0.99 \cdot 0.1 + (1 - 0.98) \cdot (1 - 0.1) = 0.117
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$$\mathsf{P}[\text{COVID} \mid \text{Positive}] = \frac{\mathsf{P}[\text{Positive} \mid \text{COVID}] \cdot \mathsf{P}[\text{COVID}]}{\mathsf{P}[\text{Positive}]} = \frac{0.99 \cdot 0.1}{0.117} = 84.6\%$$

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Bayesian statistics

In Bayesian statistics, probability expresses a degree of *belief*. How much should you believe that you have COVID-19? Answer:

$$\mathrm{P}[\mathrm{COVID}] = 10\%.$$

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Note that our belief changes as data become available.

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$$\mathsf{P}[\mathsf{COVID} \mid \mathsf{Positive}] = \frac{\mathsf{P}[\mathsf{Positive} \mid \mathsf{COVID}] \cdot \mathsf{P}[\mathsf{COVID}]}{\mathsf{P}[\mathsf{Positive}]}$$

Some terminology:

- $\mathsf{P}[\mathsf{COVID}]$ is called the *prior*
- $\mathsf{P}[\mathsf{COVID} \mid \mathsf{Positive}]$ is the *posterior*
- $\mathsf{P}[\mathsf{Positive} \mid \mathsf{COVID}]$ is the *likelihood*

These probability values are arbitrary

Bayes' Theorem

PDF version

Bayes theorem for continuous random variables

Let X and Y be two real-valued continuous random variables. Then,

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)},$$

for all y such that $p_Y(y) > 0$.

Recall that the marginal pdf, p_Y, is given by

$$p_Y(y) = \int_{-\infty}^{\infty} p_{X,Y}(x,y)\mathrm{d}x = \int_{-\infty}^{\infty} p_{Y|X}(y \mid x)p_X(x)\mathrm{d}x$$

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$$p_{X|Y}(x \mid y) = \frac{p_{Y|X}(y \mid x)p_X(x)}{p_Y(y)},$$

provided $p_Y(y) > 0$. But there's a more convenient way to write this.

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Bayes theorem for continuous random variables (better)

We can write

$$p_{X|Y}(x \mid y) \propto p_{Y|X}(y \mid x) p_X(x),$$

where the symbol $\propto$ stands for "proportional to". Here we have omitted p_Y as it does not depend on x.

- p_X: prior
- $p_{Y|X}$: likelihood
- $p_{X|Y}$: posterior

Application

Denoising

Estimation from noisy measurements

Let $X \sim \mathcal{N}(0, \sigma_X^2)$ and $N \sim \mathcal{N}(0, \sigma_N^2)$ be independent and

$$Y = X + N.$$

We measure $Y = y$. Find $p_{X|Y}(x \mid y)$.

We will use following useful lemma:

Lemma. *Let X and N be real-valued continuous random variables. Let $Y = X + N$. Then,*

$$F_{Y|X}(y \mid x) = F_N(y - x), \tag{1a}$$

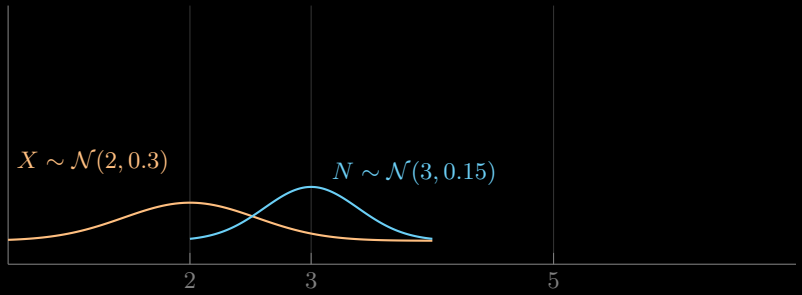
$$p_{Y|X}(y \mid x) = p_N(y - x). \tag{1b}$$

From Bayes' theorem,

$$\begin{aligned} p_{X|Y}(x \mid y) &\propto p_{Y|X}(y \mid x) p_X(x) \\ &\propto p_N(y - x) p_X(x) \\ &\propto \exp\left[-\frac{(y-x)^2}{2\sigma_N^2}\right] \exp\left[-\frac{x^2}{2\sigma_X^2}\right] \propto \exp\left[-\frac{\left(x - \frac{y\sigma_X^2}{\sigma_X^2+\sigma_N^2}\right)^2}{2\frac{\sigma_X^2\sigma_N^2}{\sigma_X^2+\sigma_N^2}}\right] = \mathcal{N}\left(\frac{y\sigma_X^2}{\sigma_X^2+\sigma_N^2}, \frac{\sigma_X^2\sigma_N^2}{\sigma_X^2+\sigma_N^2}\right). \end{aligned}$$

Estimation of X from $Y = X + N$ — Example

Suppose $X \sim \mathcal{N}(2, 0.3)$ and $N \sim \mathcal{N}(3, 0.15)$. Then, $Y \sim \mathcal{N}(5, 0.45)$.



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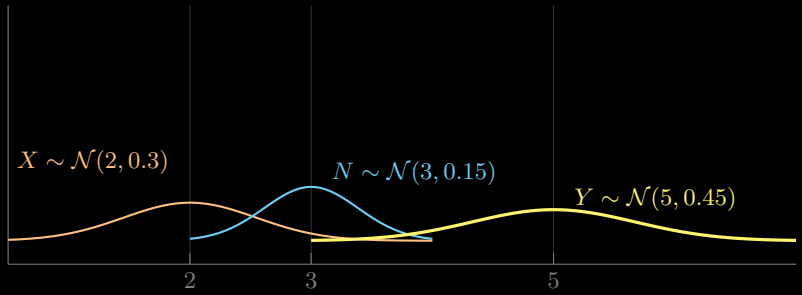

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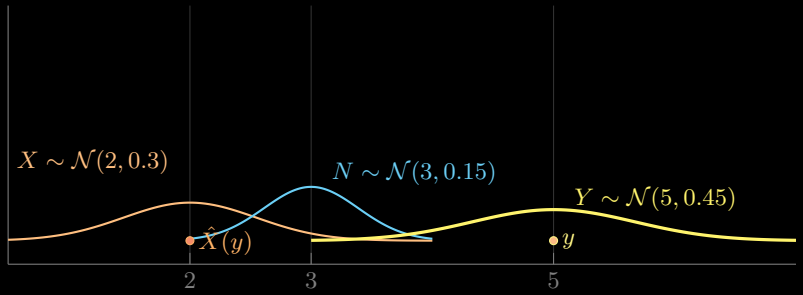

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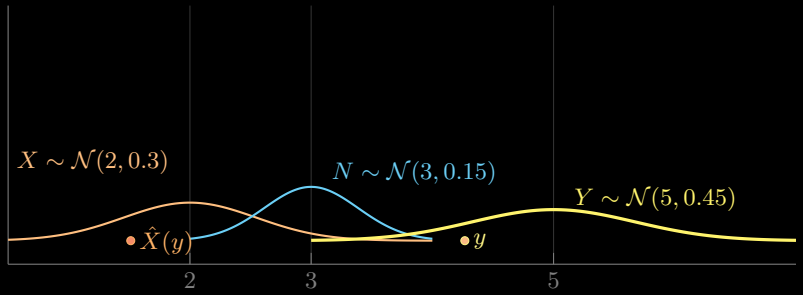

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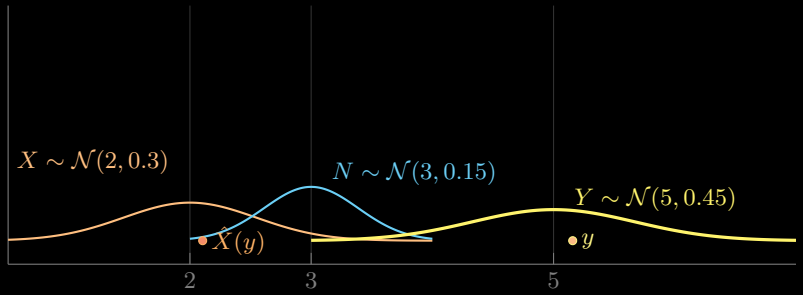

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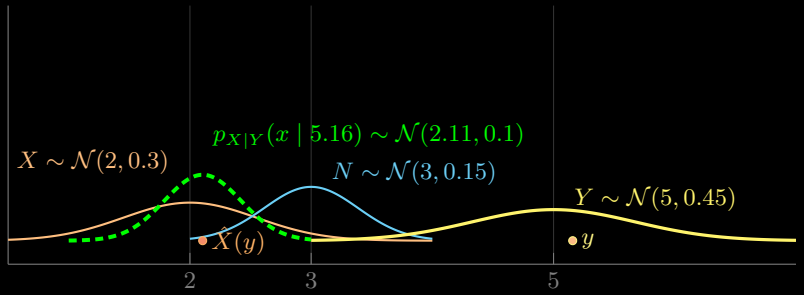

Note that $\mathrm{Var}[X \mid Y = 5.16] < \mathrm{Var}[X]$.

The posterior

We started with a prior belief about X (that is, $p_X(x)$). We obtained a measurement $Y = y$, and we determined the posterior, $p_{X|Y}(x \mid y)$.

- The posterior, $p_{X|Y}(x \mid y)$, is our updated belief/knowledge about X
- We may want to determine a single characteristic value out of $p_{X|Y}$ (a "prediction" or "estimation")
- We can determine $\mathbb{E}[X \mid Y = y] = \int p_{X|Y}(x \mid y)\mathrm{d}x$
- The conditional expectation is—in a sense—the best we can do, but can be computationally difficult
- We can determine the *mode* of $p_{X|Y}$ known as the *maximum a posteriori* estimate; this has some nice properties and is computationally easier.

Estimates and Estimators

Point estimation

Bayes' theorem gives us the posterior, $p_{X|Y}(x \mid y)$. But given an observation $Y = y$, we may want to extract a particular value $f(y)$, e.g.,

- $f_1(y) = \mathbb{E}[X \mid Y = y]$
- $f_2(y) = \text{mode}[X \mid Y = y]$
- $f_3(y) = \text{median}[X \mid Y = y]$

or any other function of y.

Estimators and estimates

Problem statement. Suppose X and Y are jointly distributed random variables. We measure Y. What is a good estimate of X given Y? We are looking for a function $\widehat{X}$ that takes an *observation* (a sample of Y) and returns an estimate of X.

$$y \longrightarrow \boxed{\widehat{X}(\,\cdot\,)} \longrightarrow \widehat{X}(y)$$

- The <u>value</u> $\widehat{X}(y)$ is an **estimate** of X given $Y = y$.
- The <u>function</u> $\widehat{X}$ is an **estimator**
- The quantity $\widehat{X}(Y)$ is a random variable
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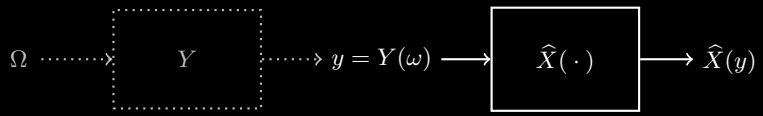

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The lowercase quantities (conditional)

Let $\widehat{X}$ be an estimator. We define

- The *(conditional) estimator error* at an observation y is

$$e(y) = X - \widehat{X}(y).$$

- The *(conditional) estimator bias at* y, is

$$\mathrm{bias}(y) = \mathbb{E}[e(y) \mid Y = y].$$

- The (conditional) expected quadratic error of the estimation error at an observation y is

$$\sigma(y) := \mathbb{E}[e(y)e(y)^\intercal \mid Y = y].$$

Note that $\sigma(y) = \mathrm{Var}[e(y) \mid Y = y]$ if $\mathbb{E}[e(y) \mid Y = y] = 0$ for all y.

The uppercase quantities (unconditional)

Let $\widehat{X}$ be an estimator. We define

- The *(unconditional) estimator error* is
$$E = X - \widehat{X}(Y).$$
- The *(unconditional) estimator bias at y*, is
$$\mathrm{Bias} = \mathbb{E}[E] = \mathbb{E}[X - \widehat{X}(Y)].$$
If $\mathrm{Bias} = 0$, we say that the estimator is **unbiased**.
- The *(unconditional) expected quadratic error* of the estimator as
$$\Sigma := \mathbb{E}[EE^\intercal].$$

Note that $\Sigma = \mathrm{Var}[E]$ if the estimator is unbiased.